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S.E. (Information Technology) (Second Semester)

EXAMINATION, 2017

PROCESSOR ARCHITECTURE AND INTERFACING

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

- N.B. :—** (i) Answer Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4,
Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.
(ii) Neat diagrams must be drawn wherever necessary.
(iii) Figures to the right indicate full marks.
(iv) Assume suitable data if necessary.

1. (a) Explain any *four* programming tools needed to write an assembly language program. [8]
(b) Draw and explain the function of control register CR0 and CR3 of 80386. [4]

Or

2. (a) What do you mean by assembler directive ? Explain the following assembler directive :
(1) DW
(2) MACRO
(3) EXTRN
(4) PUBLIC. [8]
(b) Draw the general segment descriptor format. Explain how granularity G bit affects the LIMIT Held. [4]

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3. (a) Explain page translation process with diagram. Draw PTE and PDE formats. [8]
(b) How are interrupts/exception handled in protected mode of 80386 ? Explain using various gates format. [5]

Or

4. (a) What is a Task State Segment (TSS) and how does it support Task Switching ? What is its size and content ? [8]
(b) How will you switch from real mode to protected mode and steps needed to set up the Protected mode before switching? [5]

5. (a) Draw the basic memory structure for 64K program memory and 64k data memory and 4K byte of internal program memory and 256 bytes of internal data memory. [8]
(b) Explain the following instructions of 8051 :
(1) MUL AB
(2) ADD A,#10h [4]

Or

6. (a) Draw internal memory organization of 8051. Explain its distinct advantages and disadvantage. [8]
(b) State and justify addressing mode of the following 8051 instruction :
(1) MOVX A, @DPTR
(2) MOV R6, A. [4]

7. (a) List the interrupts support by 8051 with their vector addresses. Explain interrupt programming with the help of special function register IE. [7]

- (b) Explain special function register SCON and its utility for serial port of 8051 microcontroller. [6]

Or

8. (a) Explain the following SFR of 8051 :
TMOD and TCON Register. [7]
- (b) Draw Synchronous and Asynchronous serial communication format. [6]

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